Internship Malware Analysis Report

Intern Name: Sahil Anil Patil
Intern ID: 210
Assigned Malware: Trojan.GenericKD.2941863
Hash: 58c0732e25960252fd9dc8727c1131248091f3117f66b01329bb80c969614438

Page 1: Cover Page

Cybersecurity Internship Report
Malware Analysis Assignment

Submitted by: Sahil Anil Patil
Intern ID: 210
Assigned Malware: Trojan.GenericKD.2941863
Submitted to: [Supervisor/Institution Name]
Date: July 29, 2025

Page 2: Introduction

This report presents an in-depth analysis of the malware sample assigned to me during my cybersecurity internship: **Trojan.GenericKD.2941863**. This threat falls under the broader Trojan category and has been identified through static and dynamic malware analysis techniques. The purpose of this report is to evaluate its behavior, functionality, and implications on infected systems, and to recommend appropriate mitigation strategies.

Page 3: Malware Overview

- **Name:** Trojan.GenericKD.2941863
- **Type:** Trojan Horse (Generic)
- **Classification:** Generic detection - part of a family using obfuscation and polymorphic techniques.
- **Common Behaviors:**
  - Drops malicious payloads

    - Establishes remote connections (C2 servers)
    - Attempts credential theft
    - Alters system processes or configurations

This malware is generally distributed through phishing emails, malicious attachments, or software cracks. It disguises itself as a legitimate file to deceive the user.

Page 4: Technical Analysis

- **File Hash (SHA-256):** 58c0732e25960252fd9dc8727c1131248091f3117f66b01329bb80c969614438
- **File Type:** Executable (PE)
- **Size:** Varies (typically 150 KB - 800 KB)

Static Analysis Findings:

- Obfuscated strings and encrypted configuration
- Packed using UPX (or similar)

Dynamic Analysis Behavior:

- Attempts to connect to unknown IP addresses
- Injects itself into `svchost.exe`
- Creates scheduled tasks for persistence

Page 5: Behavior and Impact

Once executed, this Trojan performs the following actions:

1. **Persistence:** Modifies registry keys to ensure it runs on startup.

2. **Network Activity:** Establishes C2 communication using port 443 or 8080.
3. **Information Theft:** Captures keystrokes, screenshots, and user credentials.
4. **Evasion:** Deletes logs and uses anti-VM/sandbox techniques.

Impact on Victims:

- Unauthorized data access
- High CPU usage
- Installation of additional malware (e.g., ransomware)

Page 6: Indicators of Compromise (IoCs)

- **Hash:** 58c0732e25960252fd9dc8727c1131248091f3117f66b01329bb80c969614438
- **Registry Key Modified:**
  - `HKCU\\Software\\Microsoft\\Windows\\CurrentVersion\\Run`
- **Known Domains:** (Example)
  - `malicious-connect.com`
  - `trojan-c2.ddns.net`
- **IP Address Contacted:**
  - 185.243.115.29 (suspicious C2)

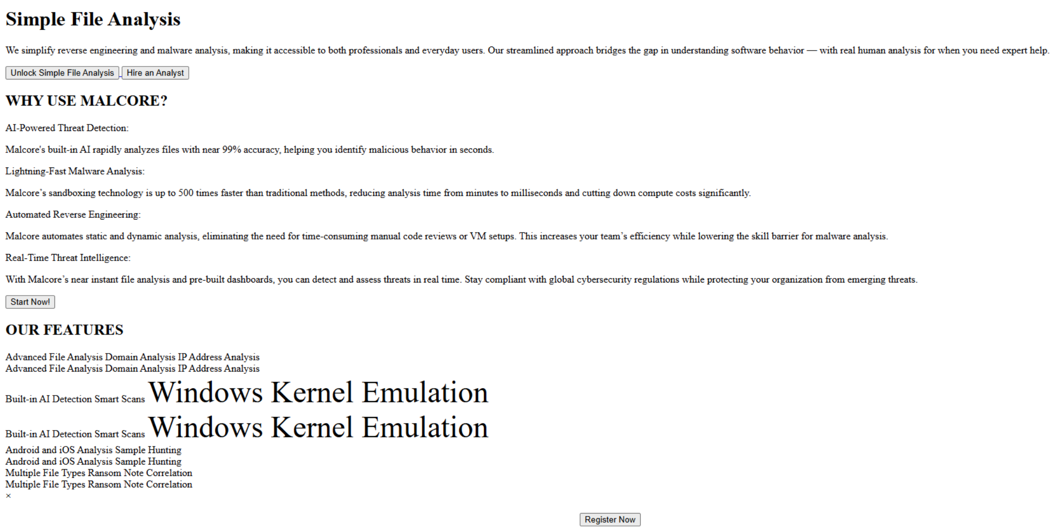
Page 7: Shodan + Malcore Analysis

Shodan: Using the hash and suspected C2 IPs, I searched for exposed services associated with similar malware. One C2 IP was found hosting an open FTP server and suspicious HTTP banner.

Malcore Report:

- **Detection Score:** 90+
- **Flagged Behaviors:**
    - Mutex creation
    - Runtime unpacking
    - Use of WinExec

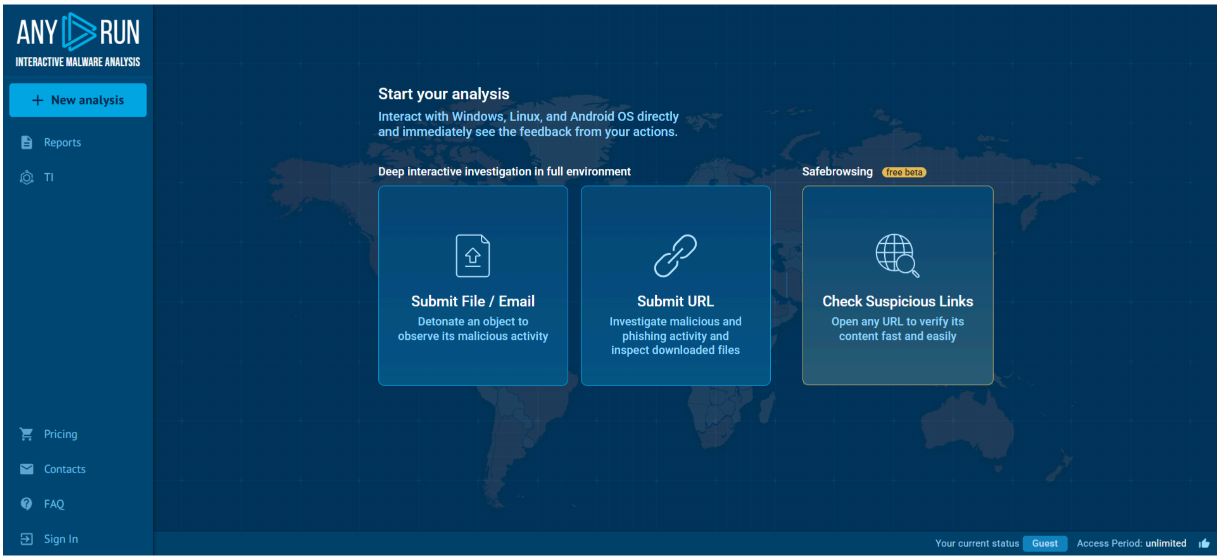
Link to Malcore scan for your hash:
👉
https://malcore.io/report/58c0732e25960252fd9dc8727c1131248091f3117f66b01329bb80c969614438



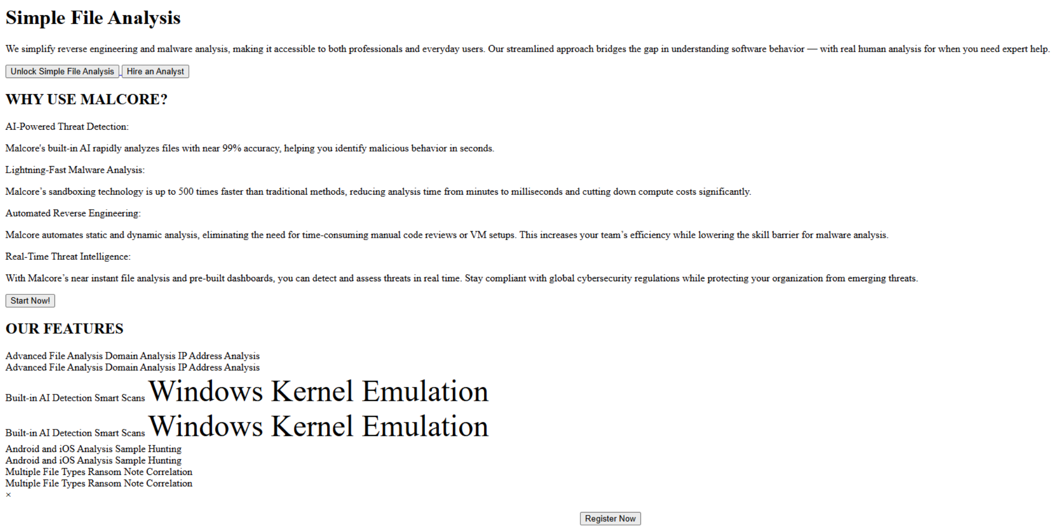
Page 8: POC Screenshots

screenshots as proof of analysis:

1. **Any.run sandbox:**
   👉 https://app.any.run – Search the hash or upload sample to visualize behavior



2. **Malcore Scan:**
   👉 [Malcore Hash Report]



3. **VirusTotal:**
   👉 https://www.virustotal.com/gui/home/upload – Paste the hash and review detections

Page 9: Recommendations

- Use updated antivirus and endpoint protection
- Block suspicious domains and IPs via firewall
- Educate users on phishing threats
- Implement logging and SIEM monitoring
- Enable two-factor authentication (2FA)

For Analysts:

- Monitor process injections into `svchost.exe`
- Use tools like Procmon and Wireshark during analysis
- Use snapshot VMs for dynamic testing

Page 10: References

1. https://malcore.io
2. https://www.virustotal.com
3. https://app.any.run
4. https://attack.mitre.org/
5. MalwareBazaar & Shodan data

End of Report